BOB WEEK

Curriculum Vitae

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EDUCATION

2020 **PhD Bioinformatics & Computational Biology** — **S.L. Nuismer Lab, IBEST, University of Idaho**
Dissertation focused on modeling eco-evolutionary processes and developing statistical methods

2015 **BS Mathematics** — **University of Idaho**
Traditional math degree with electives in electrical engineering

PEER-REVIEWED PUBLICATIONS

2026 **Stochastic Eco-Evolutionary Dynamics of Multivariate Traits** — **Journal of Theoretical Biology**
Week, B. doi:10.1016/j.jtbi.2026.112428

2025 **Quantitative Genetics of Microbiome Mediated Traits** — **Evolution**
Week, B.; Ralph, P.L.; Tavalire, H.F.; Cresko, W.A.; Bohannan, B.J.M. doi:10.1093/evolut/qpaf171

2025 **Applying Evolutionary Theory to Understand Host-Microbiome Evolution** — **Nature EcoEvo**
Week, B.; Russell, S.L.; Schulenburg, H.; Bohannan, B.J.M.; Bruijning, M. doi:10.1038/s41559-025-02846-w

2023 **Host-Parasite Coevolution in Continuous Space Leads to Variation in Local Adaptation Across Spatial Scales** — **The American Naturalist**
Week, B.; Bradburd, G.S. doi:10.1086/727470

2022 **Uncovering Cryptic Coevolution** — **The American Naturalist**
Nuismer, S.L.; *Week, B.*; Harmon, L.J. doi:10.1086/717436

2021 **A White Noise Approach to Evolutionary Ecology** — **Journal of Theoretical Biology**
Week, B.; Nuismer, S.L.; Harmon, L.J.; Krone, S.M. doi:10.1016/j.jtbi.2021.110660

2021 **Coevolutionary Arms Races and the Conditions for the Maintenance of Mutualism** — **The American Naturalist**
Week, B.; Nuismer, S.L. doi:10.1086/714274

2021 **A Unified Model of Species Abundance, Genetic Diversity, and Functional Diversity Reveals the Mechanisms Structuring Ecological Communities** — **Molecular Ecology Resources**
Overcast, I.; Ruffley, M.; Rosindell, J.; Harmon, L.; Borges, P.; Emerson, B.; Etienne, R.S.; Gillespie, R.; Krehenwinkel, H.; Mahler, L.; Massol, F.; Parent, K.; Patiño, J.; Peter, B.; *Week, B.*; Wagner, C.; Hickerson, M.J.; Rominger, A. doi:10.1111/1755-0998.13514

2019 **Identifying Models of Trait-Mediated Community Assembly using Random Forests and Approximate Bayesian Computation** — **Ecology and Evolution**
Ruffley, M.; Peterson, K.; *Week, B.*; Tank, D.; Harmon, L.J. doi:10.1002/ece3.5773

2019 **Approximate Bayesian Estimation of Coevolutionary Arms Races** — **PLOS Computational Biology**
Nuismer, S.L.; *Week, B.* doi:10.1371/journal.pcbi.1006988

2019 **The Measurement of Coevolution in the Wild** — **Ecology Letters**
Week, B.; Nuismer, S.L. doi:10.1111/ele.13231

2018 **Coevolution Slows the Disassembly of Mutualistic Communities** — **The American Naturalist**
Nuismer, S.L.; *Week, B.*; Aizen, M. doi:10.1086/699218

PREPRINTS

2024 **The Evolution of Microbiome-Mediated Traits** — **bioRxiv**
Week, B.; Morris, A.H.; Bohannan, B.J.M. doi:10.1101/2024.03.29.587374

MANUSCRIPTS IN PREPARATION

2026 **Microbially Assisted Rescue of Host Populations** — **Manuscript in preparation**
Week, B.; Schulenburg, H.; Uecker, H. doi:

AWARDS

2024 – 2027 **EU/DFG Postdoctoral Fellowship** — **KiTE, Kiel University**
The "Kiel Training for Excellence" programme is cofunded by the Marie Sklodowska-Curie Actions from the European Commission's Horizon Europe programme (project number: 101081480) and by Kiel University

2018 – 2019 **Bioinformatics & Computational Biology Fellowship** — **IBEST, University of Idaho**
Project aimed to model the duration of coevolutionary associations

2017-2018 **Bioinformatics & Computational Biology Fellowship** — **IBEST, University of Idaho**
Project aimed to develop a statistical method to measure coevolution in continuous space

2017 **Paul Joyce Memorial BCB Fellowship Endowment** — **IBEST, University of Idaho**
Nominated by Professor Scott Nuismer because of my "love for mathematics and helping others to appreciate how it can be used to understand biological processes"

2014-2015 **Undergraduate Research in Biology & Mathematics** — **IBEST, University of Idaho**
Efforts focused on developing a statistical method to measure coevolution in metapopulations

PROFESSIONAL EXPERIENCE

2024 – 2027 **KiTE Postdoctoral Research Fellow** — **H. Schulenburg Lab, Kiel University**
Establishing theoretical foundations for the study of microbiome mediated trait dynamics

2024 – Present **Guest Researcher** — **Max Planck Institute for Evolutionary Biology, Plön**
Concurrent research affiliation providing access to MPI research facilities and scientific community

2022 – 2024 **Postdoctoral Research Fellow** — **B.J.M. Bohannan Lab, University of Oregon**
Extended evolutionary theory for traits jointly determined by host genotype and host microbiome

2020 – 2022 **Postdoctoral Researcher** — **G.S. Bradburd Lab, Michigan State University**
Developed mathematical and computational approaches to understand coevolution in continuous space

2018 **Visiting Scientist** — **P.J. CaraDonna Lab, Rocky Mountain Biological Laboratory**
Field ecology training on estimating floral abundance and phenology, recording plant-pollinator interactions and estimating percent cover

TEACHING EXPERIENCE

2026 **Guest Lecturer** — **Kiel University, BIOL227 - Ecology, Evolution, and Genetics**
Introduced coevolution to a masters student class, emphasizing ecological aspects and touching on the role of geography.

2017 **Teaching Assistant** — **University of Idaho, Department of Biological Sciences**
Taught the lab portion of a 300-level ecology and population biology course

2011 – 2013 **Mathematics Tutor** — **Clark Community College, Mathematics Department**
Part-time work at tutoring center supporting students taking a wide-range of coursework

PRESENTATIONS

2026 **Charting the Evolutionary Consequences of Interspecific Interactions** - Invited Seminar — **Laboratoire de Biométrie et Biologie Évolutive (LBBE) - Lyon, France**

2026 **The Response of Genetic Correlations to Drift** - Talk — **Aquavit Seminar - MPI, Plön**

2026 **Microbially Assisted Rescue of Host Populations** - Poster — **ISEMPH 2026 - Kiel, Germany**

2026 **The Response of G-matrices to Random Genetic Drift** - Talk — **MMEE 2026 - Cork, Ireland**

2026 **Modelling Host-Microbiome Evolution** - Seminar — **Washington State University Vancouver - Vancouver, Washington**

2026 **Modelling Host-Microbiome Evolution** - Seminar — **University of California, Davis - Davis, California**

2026 **Modelling Host-Microbiome Evolution** - Seminar — **University of Idaho - Moscow, Idaho**

2026 **Modelling Host-Microbiome Evolution** - Seminar — **University of Oregon - Eugene, Oregon**

2026 **Quantitative Genetics of Microbiome Mediated Traits** - Talk — **BIRS - Banff, Canada**

2025	**Quantitative Genetics of Microbiome Mediated Traits** - Talk	**ESEB 2025 - Barcelona, Spain**
2025	**Quantitative Genetics of Microbiome Mediated Traits** - Seminar	**Theoretical Biology Department - MPI, Plön**
2025	**When is Microbial Rescue More or Less Effective than Genetic Rescue?** - Poster	**Evolutionary Rescue Workshop - MPI, Plön**
2025	**Microbiome-Mediated Host Adaptation: A Niche Construction Approach** - Poster	**Concepts in Evolution Workshop - MPI, Plön**
2024	**Host-Parasite Coevolution & Microbiome-Mediated Adaptation** - Seminar	**TransEvo Core Seminar - CAU, Kiel**
2023	**The Evolution of Microbiome-Mediated Traits** - Talk	**Symbiosis Theory Workshop - Eugene, Oregon**
2023	**Modeling Adaptation of Microbiome-Mediated Traits** - Talk	**EvoWibo - Port Townsend, Washington**
2022	**Host-Parasite Coevolution in Continuous Space** - Poster	**PEQG2022 - Pacific Grove, California**
2021	**Coevolutionary Arms Races and The Conditions for The Maintenance of Mutualism** - Talk	**AmNat2021 - Virtual**
2020	**A Bayesian Methodology for Estimating the Distribution of Coevolution within Ecological Communities** - Talk	**AmNat2020 - Pacific Grove, California**
2018	**The Measurement of Coevolution in Nature** - Poster	**EvoWibo - Port Townsend, Washington**
2017	**The Measurement of Coevolution in Mutualisms** - Talk	**Evolution - Portland, Oregon**

SERVICE & LEADERSHIP

2022	**Code Contributer** Developed a nucleotide-based model of coevolution for SLiM. See §19.7 *here*.	**SLiM 4.0** doi:10.1086/723601
2018-2019	**Graduate Student Representative** Represented graduate students in the Bioinformatics & Computational Biology program at institutional meetings	**IBEST, University of Idaho**
——	**Manuscript Reviewer** The American Naturalist, Ecology, Evolution, Journal of Evolutionary Biology, PCI Evol Biol, Population Ecology, Proceedings of The Royal Society B, Theoretical Population Biology, Molecular Biology & Evolution	

SOCIETIES

2021-Present	**The International Society of Nonbinary Scientists**	**isnbs.org**
2020-Present	**The American Society of Naturalists**	**amnat.org**

INTERESTS

I am broadly interested in collaborating on any scientific topic where my skills are useful. I am particularly interested in developing and formalizing models to clarify conceptual issues in population biology and community ecology.

SKILLS

Software: LaTeX, Python, R, Linux, Julia, Mathematica, SLURM, SLiM, C/C++

Statistics: Modeling, Analysis, Inference, Methods Development

Math: Linear Algebra, Dynamical Systems, Functional Analysis, Stochastic Processes